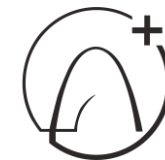




Agent Based Modelling

An Observation Journey

Morphotect



دانشگاه علم و صنعت ایران

This is Us!



Kiana Jalilian

Bachelor of Architecture



Zahra Shahriar

Master of Architecture



Amir Hossein Panahi

Master of Architecture

Welcome To Our Exploration Journey!

Our Journey: A Quick Overview

01

Study Subject Design

Overview of the subject chosen for analysis

02

Problem-Solving Approach

Proposed methodology for tackling the issue

03

Algorithm & Code Structure

Logic behind the solution and pseudo-code

04

Output Display

Visual results of the algorithm

05

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Real-world uses of the algorithm in design and planning

06

Project Challenges

Limitations and difficulties faced during problem-solving

07

Future Research Suggestions

Recommendations for advancing the project in future studies



1

Study Subject Design

We are Observers, Right?

As human beings, we are fundamentally **observers** of the world around us. Our role as observers shapes how we understand, interact with, and influence our surroundings.

Why we do this?

Capturing the feelings of an agent without directly asking questions

Capturing the feelings of an agent without directly asking questions is crucial for gaining authentic, real-time emotional insights while minimizing intrusion. Direct questioning can lead to biased or incomplete responses, as individuals may not always express their true emotions due to social pressure or lack of self-awareness. By using behavioral, physiological, and contextual data, emotions can be monitored continuously and non-invasively, offering more accurate reflections of emotional states. This approach enables real-time detection of emotions, allowing for immediate interventions in contexts such as customer service, healthcare, or workplace well-being. Ultimately, this method enhances decision-making by providing richer, unbiased emotional data without disrupting the agent's experience.

Inspiration?

Imagine you are Gil Pender and you are going to spend a Midnight in Paris!



What we look for?

The visual experience of a user in an urban environment

The visual experience of a user in an urban environment is a multi-dimensional phenomenon that combines perceptual, cognitive, and emotional elements. Research in this domain draws from urban design, environmental psychology, architecture, and cognitive science to understand how users perceive and interact with urban spaces.



A minimalist illustration on a light gray background. A winding, double-lined path in a light gray color meanders across the frame. A small, dark silhouette of a person is walking along the path, positioned towards the upper right. The text "What we see as an observer?" is written in a bold, dark blue font, centered horizontally and partially overlaid by the path.

What we see as an observer?

01

Architectural Diversity
Variety in Architectural Styles

02

Natural Landscapes and Green Spaces
Parks and Trees

03

Lines of Sight and Perspective
Urban Layout

04

Light and Shadow
Impact of Natural Light

05

Human-Scale Spaces
Proportion and Spatial Size

Visual experiences in an urban pathway

10

Signs and Urban Symbols
Directional and Advertising Signs

06

Gathering Spaces
Public Squares and Meeting Points

07

Dynamic Visual Elements
Movement of People and Vehicles

08

Interaction with Buildings and Shops
Storefronts and Window Displays

09

Art and Sculptures
Public Art Installations

What we feel as an observer?





Sense of Safety and Security

Lighting and Visibility
Crowding and Isolation



Sense of Belonging and Place Attachment

Familiarity and Identity
Social Connections



Aesthetic Appreciation and Beauty

Order and Coherence
Visual Appeal



Nostalgia and Memory

Historical and Cultural Significance
Personal Experiences



Stress and Overstimulation

Noise and Sensory Overload
Escape and Refuge



Control and Autonomy

Freedom of Movement
Choice and Flexibility



Fear and Anxiety

Social and Environmental Threats
Unfamiliarity



Surprise and Discovery

Serendipity
Exploration and Playfulness



Joy and Happiness

Vibrancy and Liveliness
Green Spaces and Nature

Emotional factors in urban experience

What if we could do something about it?

A minimalist illustration of a person walking on a winding path. The path is composed of two parallel, light gray lines that curve and loop across the white background. A small, dark silhouette of a person is walking along the path, positioned towards the upper right. The text is centered in the middle of the image, overlaid on the path.

Can we enhance the quality of visual experiences as an architect and make the agent feel better?

2

Problem-Solving Approach

But, How?



First.

We found a place in real world!

The exact location of midnight walking of Gil Pender



Where?



85-75 Rue Galande, 75005 Paris, France
The address **85-75 Rue Galande, 75005 Paris, France** refers to a street in the Latin Quarter, a historic area in the 5th arrondissement of Paris. This neighborhood is known for its rich cultural heritage, medieval architecture, vibrant cafés, and artistic charm. The Rue Galande is close to landmarks like the Notre-Dame Cathedral and the Seine River, making it a popular spot for tourists. This area has a strong connection to Parisian intellectual and artistic life, often frequented by students, writers, and creatives throughout history.

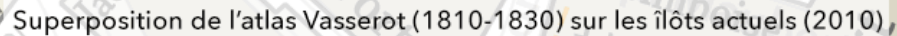
Second.

In the next step, the user specifies the origin and destination, and we guide the user along the path.

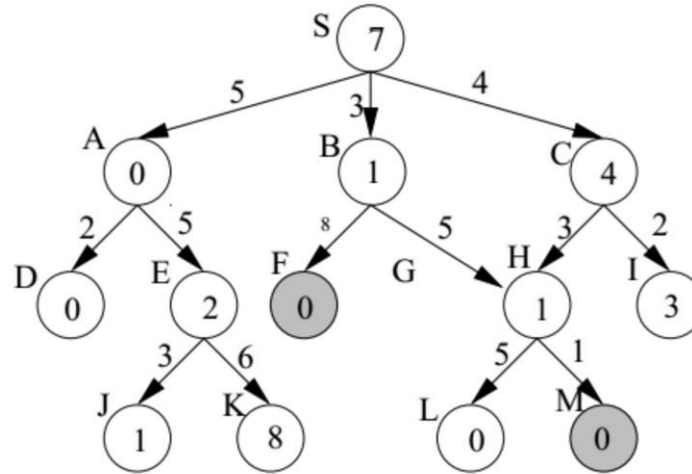


A small silhouette of a person walking on a winding, double-lined path that snakes across the upper half of the image. The path starts from the bottom left, loops around, and continues towards the top right. The person is positioned on the upper curve of the path, facing right.

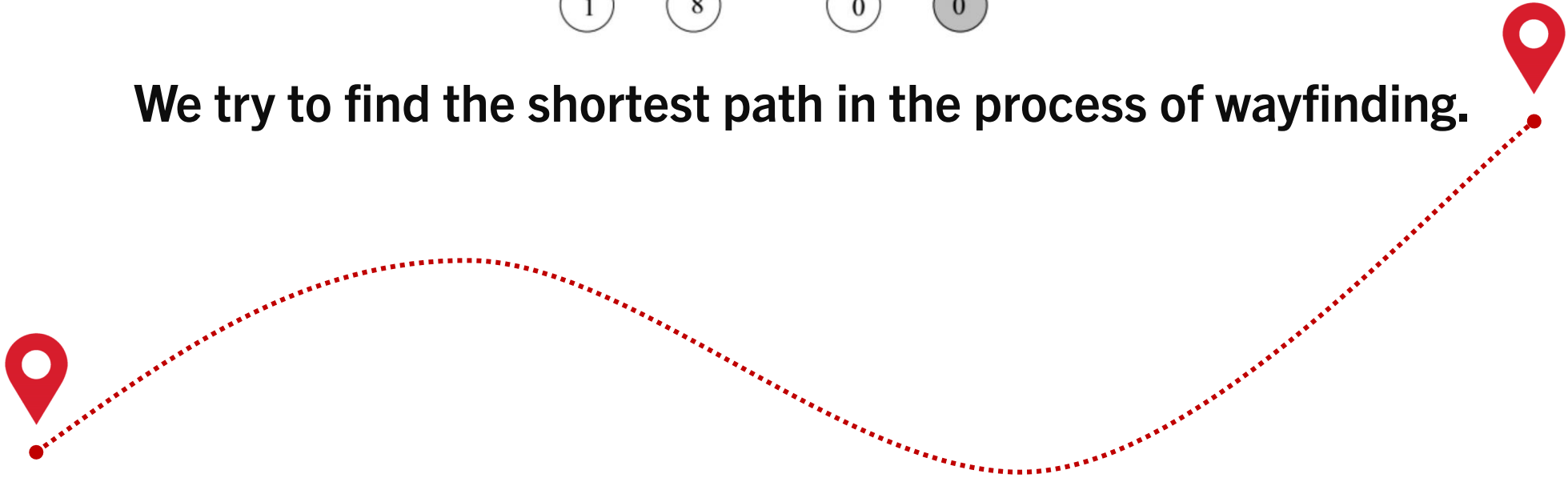
Our agent's journey starts from here!



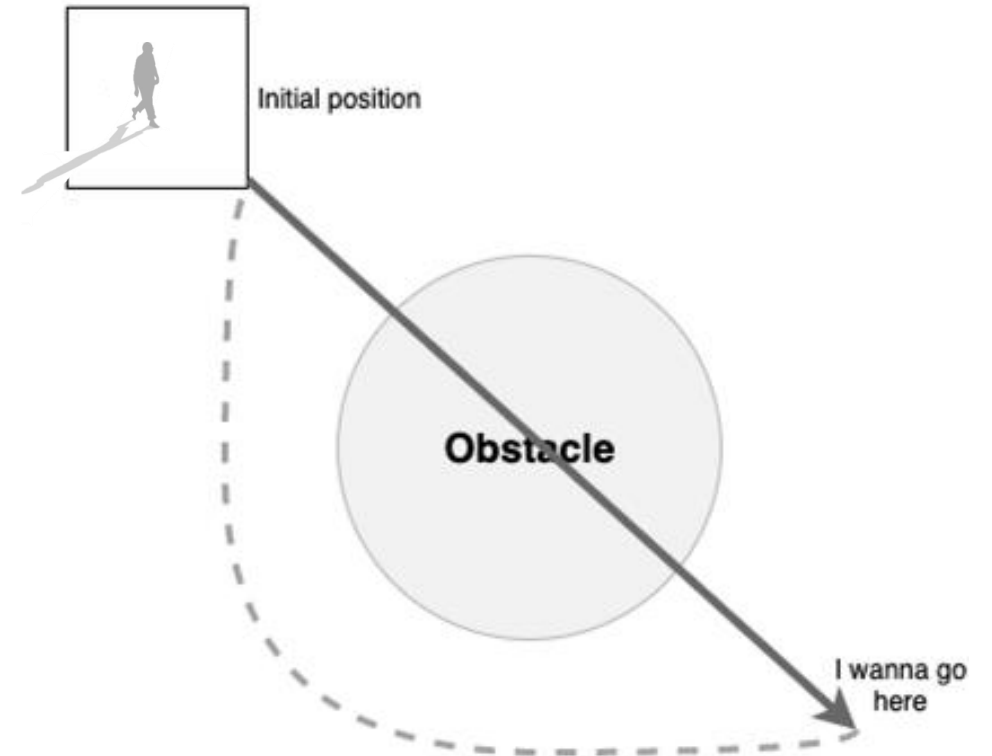
Third.



We try to find the shortest path in the process of wayfinding.



Forth.



**Of course, we were aware of obstacles
and collision avoidance!**

A small silhouette of a person walking on a winding, light gray path that curves across the upper right and middle of the slide. The person is walking away from the viewer, and a long, dark shadow is cast behind them on the path.

Fifth.

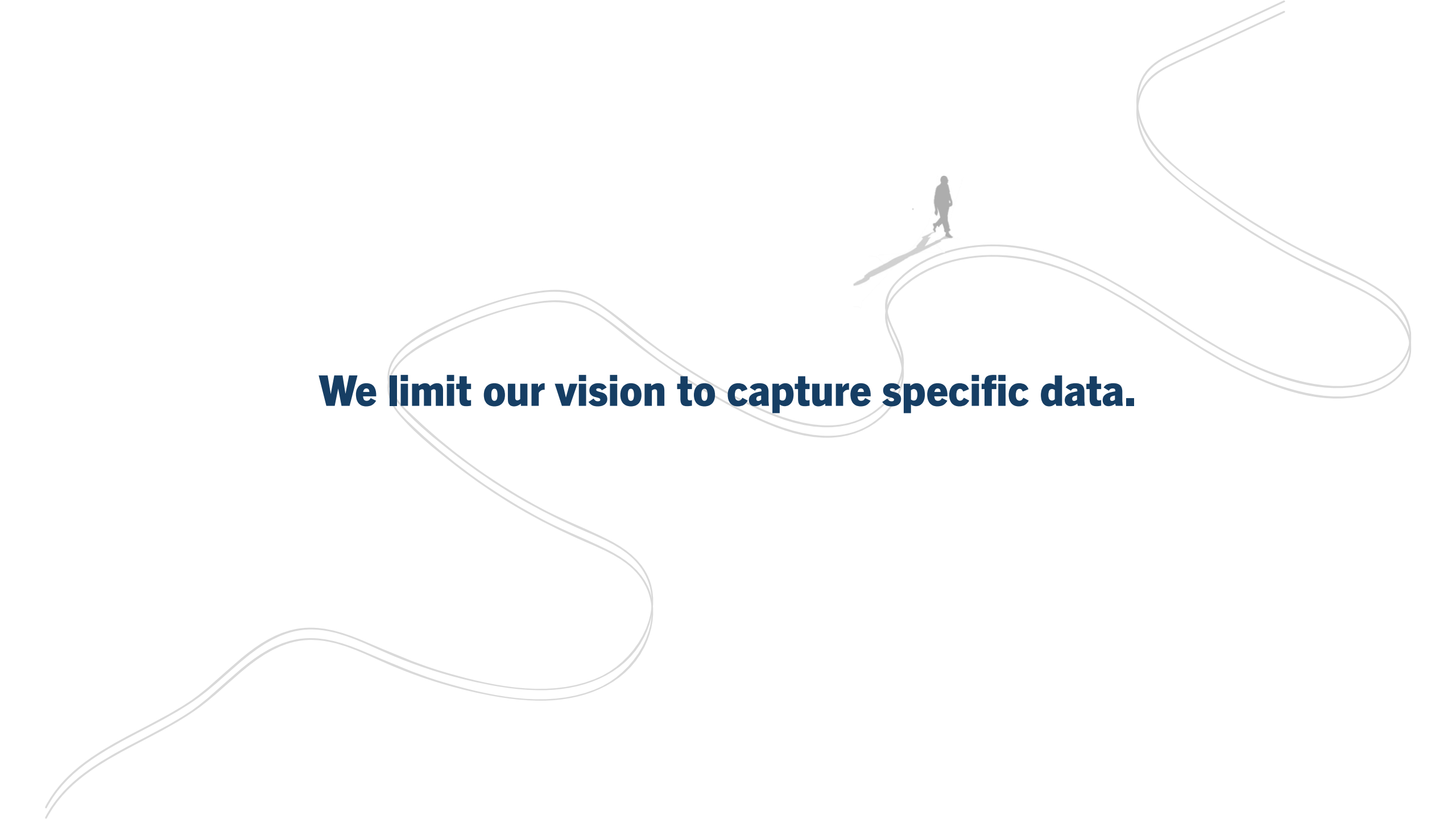
We use computational geometry and mathematical tools to identify our relationships.

By comparing areas and determining the average distances to obstacles at each point along the path, we obtain data that allows us to establish meaningful connections for determining indices, which will be detailed in the next chapter.

Then.

Finding and extracting values and indices



A minimalist illustration on a light gray background. A thin, double-lined path winds across the frame in a series of loops and curves. A small, dark silhouette of a person is walking along the path, positioned slightly to the right of the center. The text "We limit our vision to capture specific data." is written in a bold, dark blue font, centered horizontally and partially overlaid by the winding path.

We limit our vision to capture specific data.

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Architectural Diversity
Variety in Architectural Styles

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Sense of Safety and Security

Lighting and Visibility
Crowding and Isolation



Stress and Overstimulation

Noise and Sensory Overload
Escape and Refuge



Control and Autonomy

Freedom of Movement
Choice and Flexibility



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Familiarity and Identity
Social Connections



Fear and Anxiety

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Unfamiliarity



Aesthetic Appreciation and Beauty

Order and Coherence
Visual Appeal



Surprise and Discovery

Serendipity
Exploration and Playfulness



Nostalgia and Memory

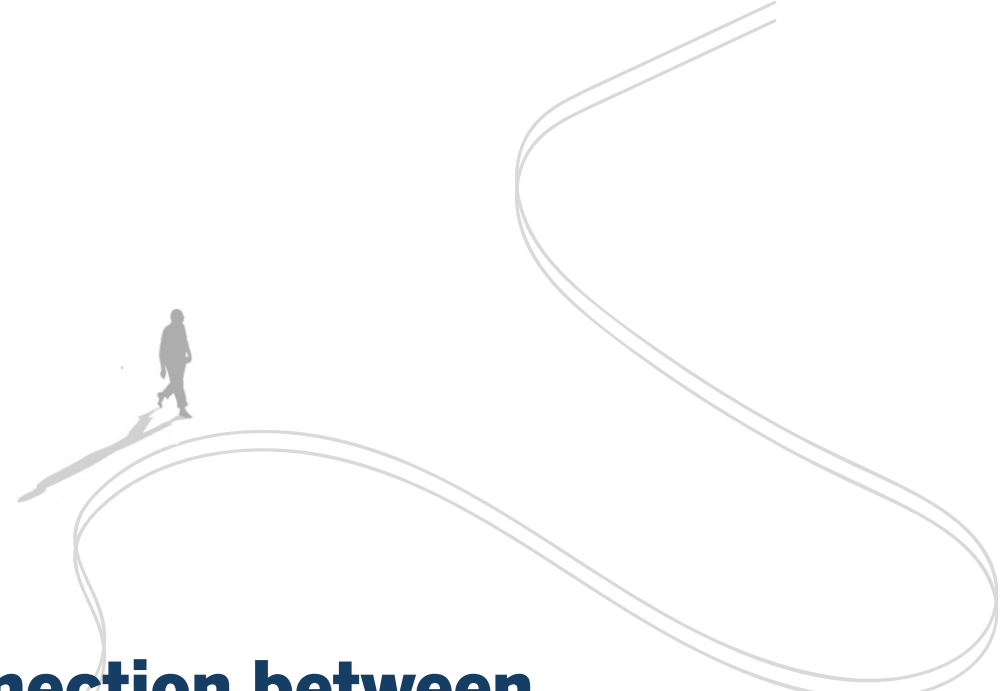
Historical and Cultural Significance
Personal Experiences



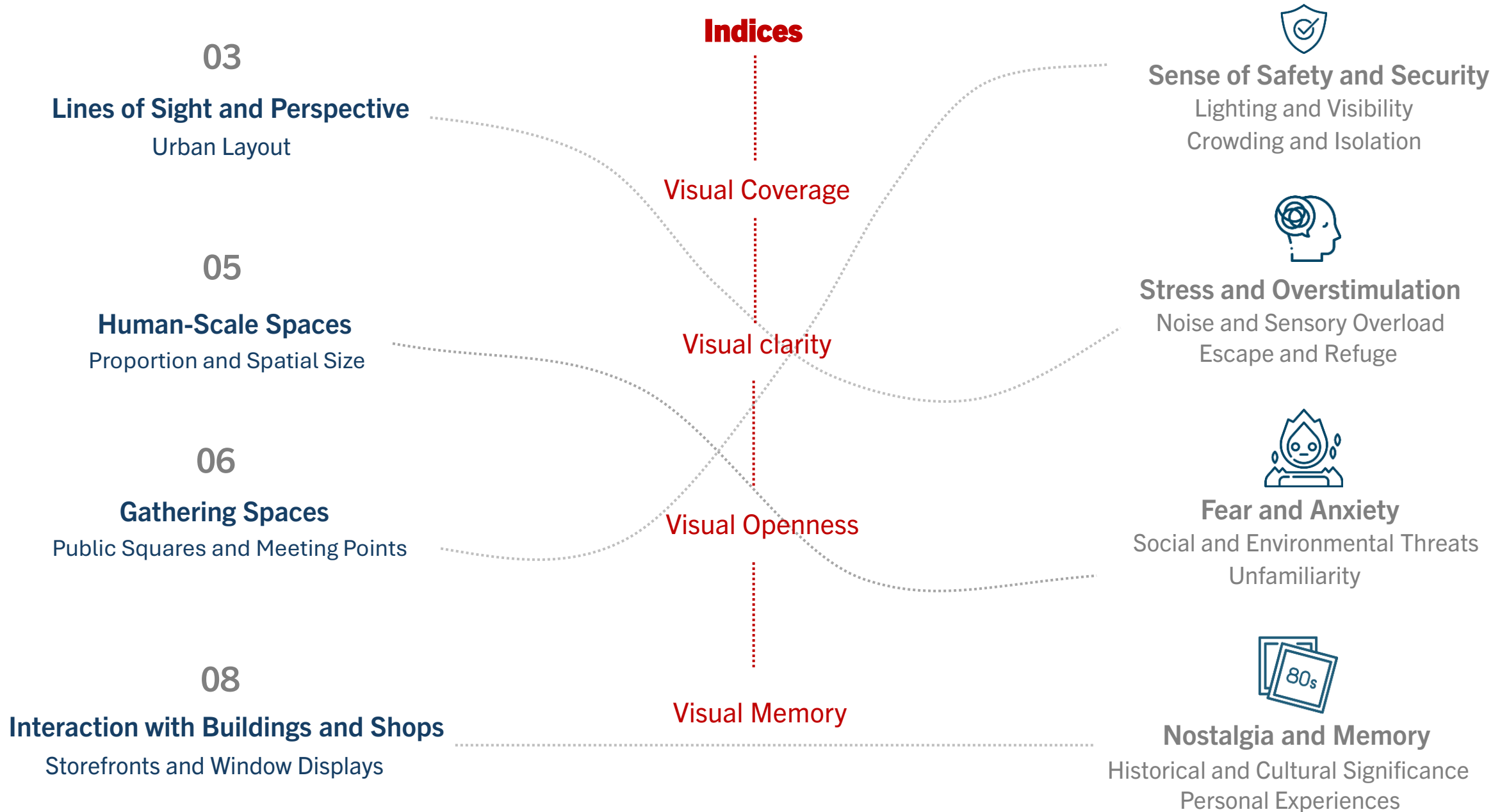
Joy and Happiness

Vibrancy and Liveliness
Green Spaces and Nature

Emotional factors in urban experience

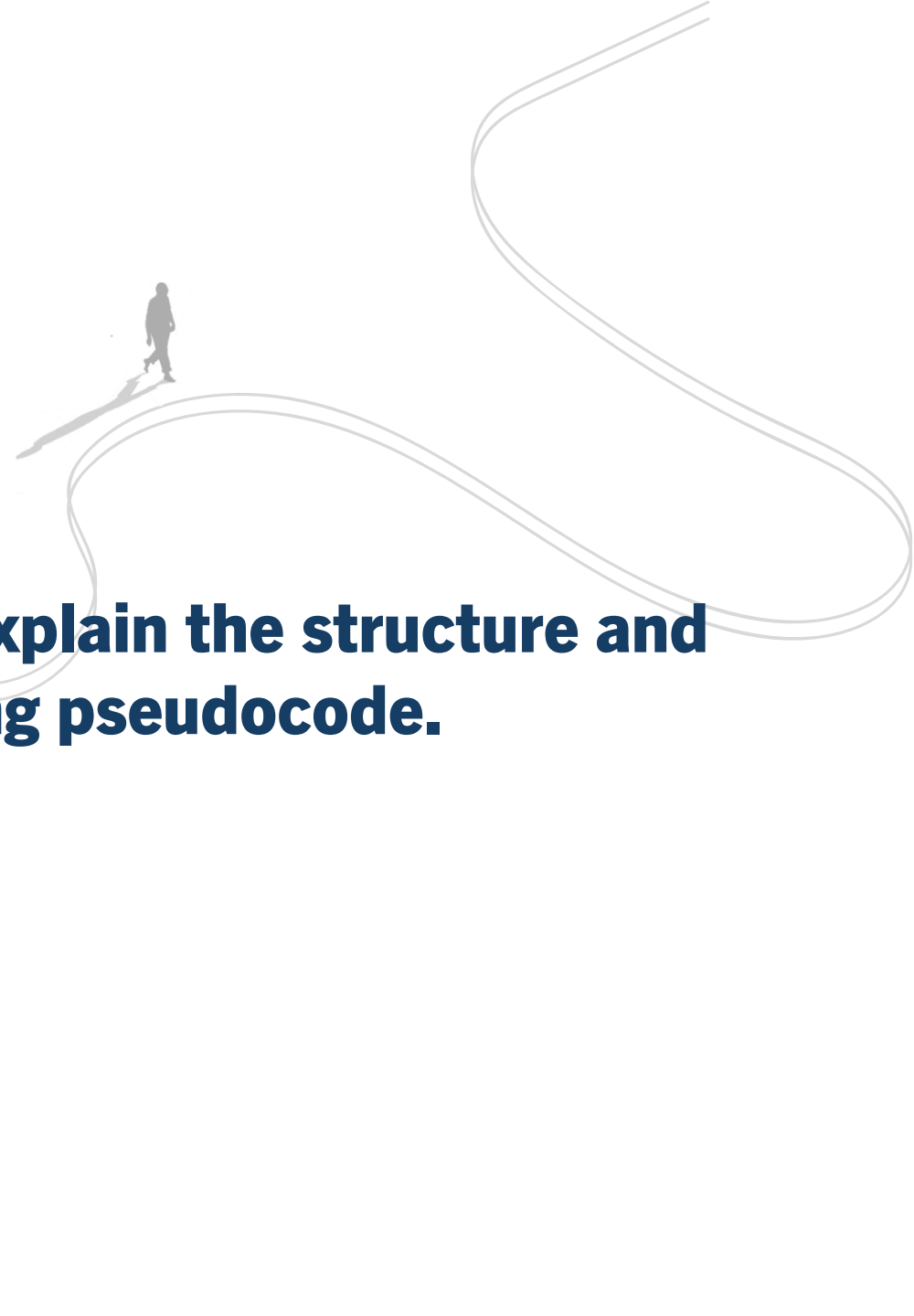
A minimalist illustration of a person walking on a winding path. The path is composed of two parallel, light gray lines that curve and loop across the upper right portion of the image. The person is a small silhouette walking away from the viewer, with a long shadow cast behind them. The background is a solid light gray.

**We aimed to establish a connection between
vision and emotion to develop meaningful indices.**



3

Algorithm & Code Structure

A minimalist illustration of a person walking on a winding, double-lined path that snakes across the frame. The path is light gray, and the person is a small silhouette. The background is white.

In this part of our journey, we will explain the structure and logic of the algorithms using pseudocode.

Here are some important ones!



A* algorithm For Path finding



Collision Avoidance



Globalization



Areas



2D isovist intersects

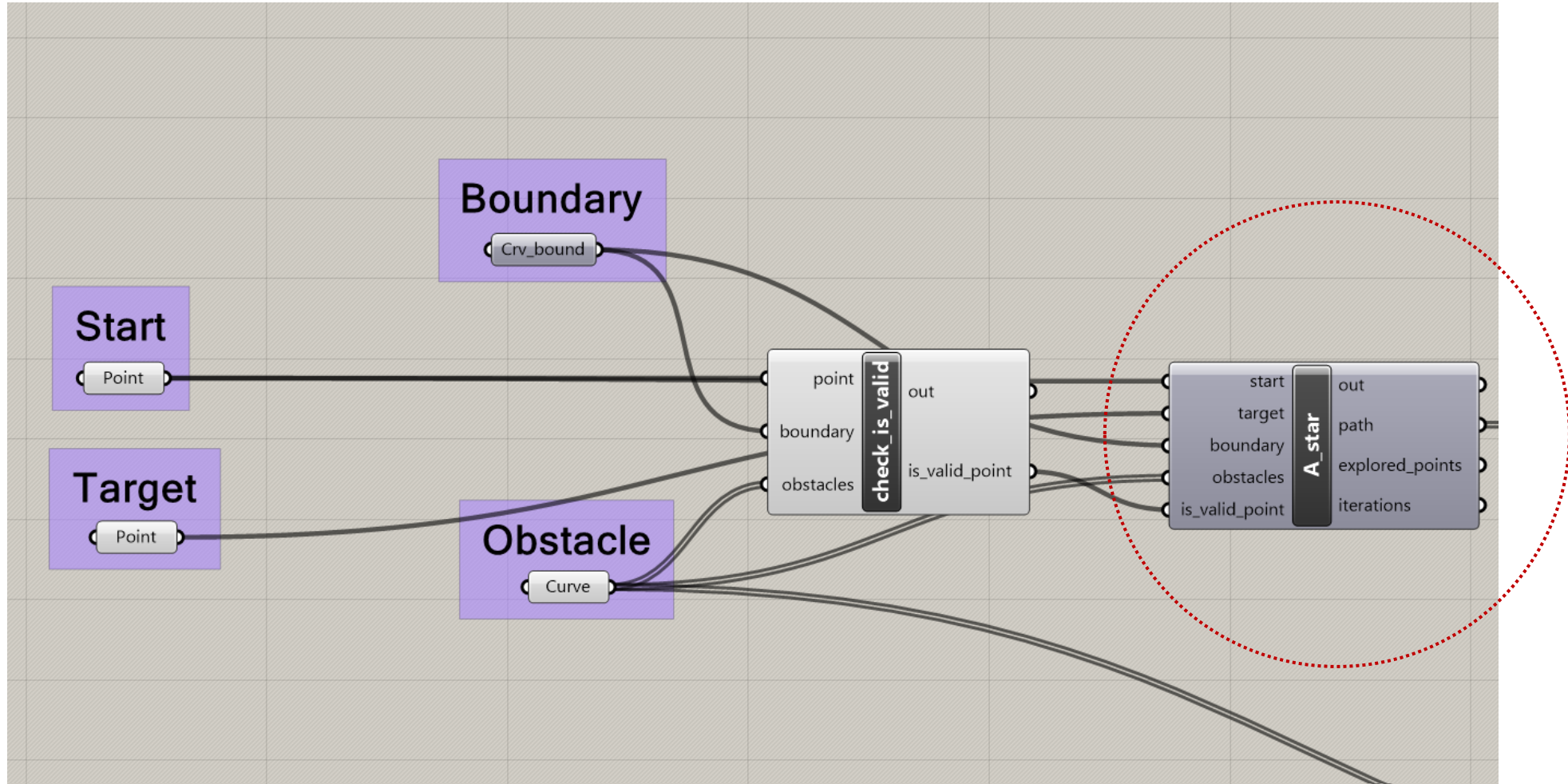


Simulation





A* algorithm For Path finding





A* algorithm For Path finding

1. Heuristic Function:

- The heuristic function calculates the Manhattan distance between the current point and the target point. This is used as the heuristic score in the A* algorithm.
- The function extracts the x, y, and z coordinates of the point and the target, calculates the absolute difference between them, and sums the differences to get the Manhattan distance.

2. A* Shortest Path Algorithm:

- The `a_star_shortest_path` function implements the A* algorithm to find the shortest path between the start and target points, considering the boundary and obstacles.
- It initializes the frontier list with the start point, the visited list, and the explored points list.
- The main loop of the algorithm involves:
 - Sorting the frontier list by the `f_score` (`g_score` + heuristic).
 - Popping the node with the lowest `f_score` from the frontier list.
 - Checking if the current position is close enough to the target.
 - Marking the current position as visited if it hasn't been visited before.
 - Exploring all possible movements from the current position.
 - Calculating the `g_score` and `f_score` for the next position.
 - Checking if the next position is valid (within the boundary and not in an obstacle).
 - Adding the valid and unvisited next position to the frontier list.
 - Appending the new points to the explored points list.
- If the target is reached, the function returns the path, explored points, and iteration count.
- If no path is found, the function returns None for the path, the explored points list, and the iteration count.



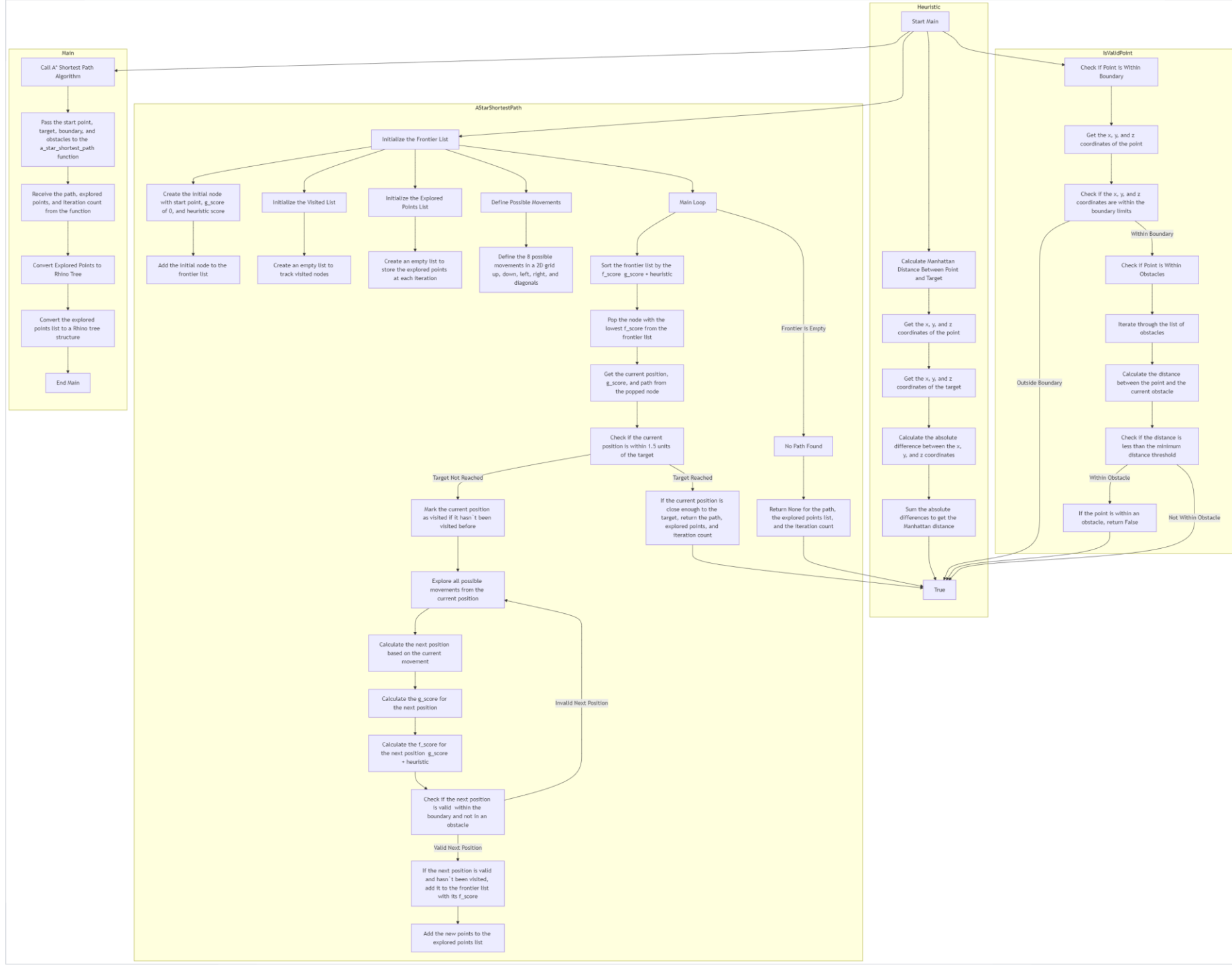
A* algorithm For Path finding

3. Is Valid Point:

- The `is_valid_point` function checks if a given point is within the boundary and not within any obstacles.
- It first checks if the point's x, y, and z coordinates are within the boundary limits.
- It then iterates through the list of obstacles and calculates the distance between the point and each obstacle. If the distance is less than the minimum distance threshold, the point is considered to be within an obstacle, and the function returns `False`.
- If the point is within the boundary and not within any obstacles, the function returns `True`.

4. Main:

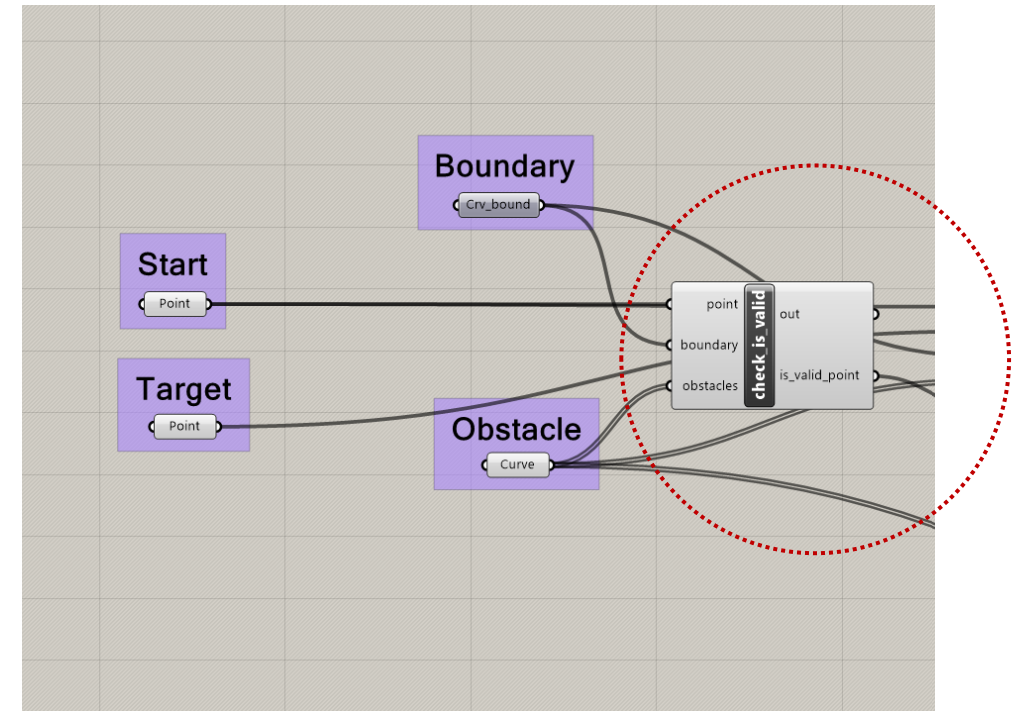
- The main part of the code calls the `a_star_shortest_path` function with the start point, target, boundary, and obstacles.
- It receives the path, explored points, and iteration count from the function.
- The explored points list is then converted to a Rhino tree structure using the `list_to_tree` function from the `ghpythonlib.treehelpers` module.



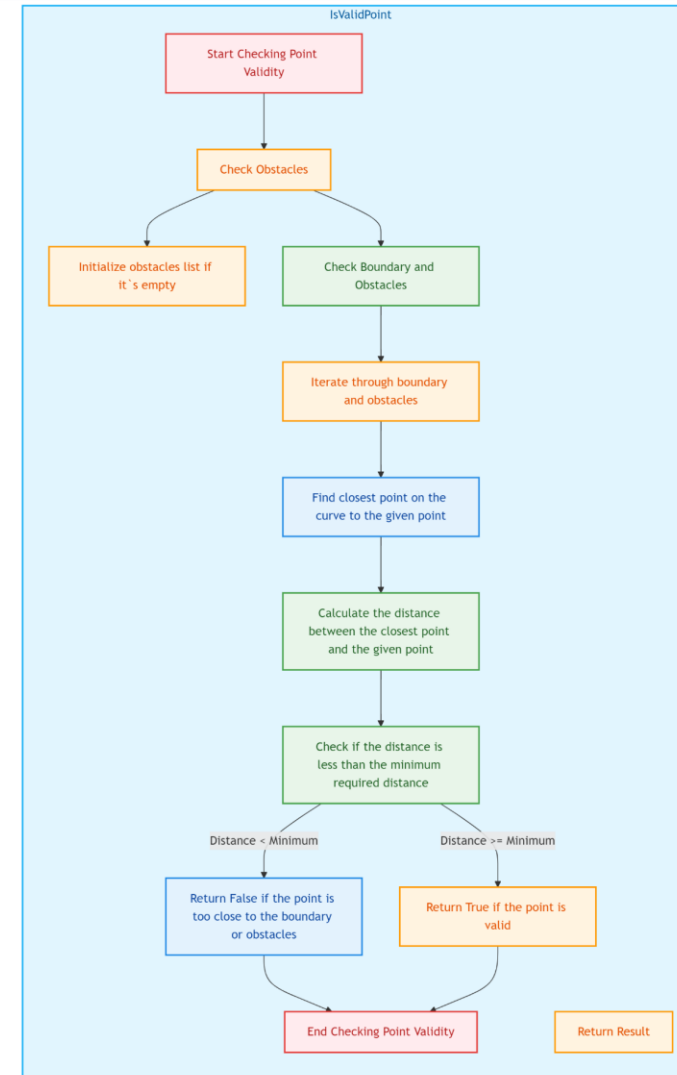
Collision Avoidance

Code Explanation:

1. The `is_valid_point` function checks if a given point is valid based on a boundary and a list of obstacles. The function takes four parameters: `point`, `boundary`, `obstacles`, and `min_dist`.
2. If the obstacles list is empty, it is initialized as an empty list.
3. The function iterates through the boundary and obstacles curves, finding the closest point on each curve to the given point. It calculates the distance between the closest point and the point.
4. If the distance is less than the `min_dist` parameter, the function returns `False`, indicating that the point is not valid.
5. If the point is valid (i.e., the distance to all boundary and obstacles is greater than or equal to the `min_dist`), the function returns `True`.



Collision Avoidance



Checks

Code

The Unsupported markdown: codespan function checks if a given point is valid based on a boundary and a list of obstacles.

If the Unsupported markdown: codespan list is empty, it is initialized as an empty list.

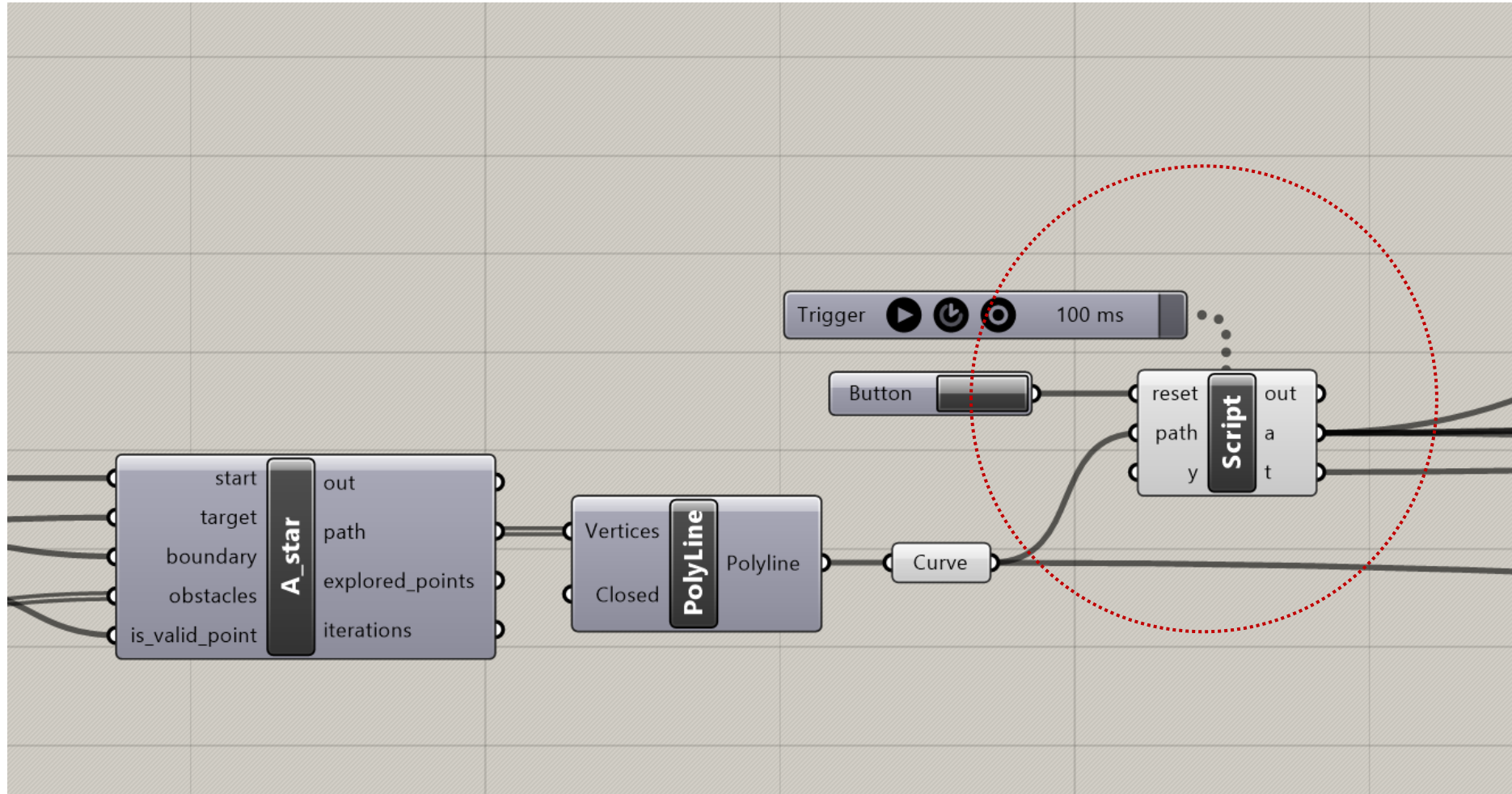
The function iterates through the Unsupported markdown: codespan and Unsupported markdown: codespan curves, finding the closest point on each curve to the given Unsupported markdown: codespan.

The distance between the closest point and the Unsupported markdown: codespan is calculated. If the distance is less than the Unsupported markdown: codespan parameter, the function returns Unsupported markdown: codespan, indicating that the point is not valid.

If the point is valid, i.e., the distance to all boundary and obstacles is greater than or equal to the Unsupported markdown: codespan, the function returns Unsupported markdown: codespan.



Globalization





Globalization

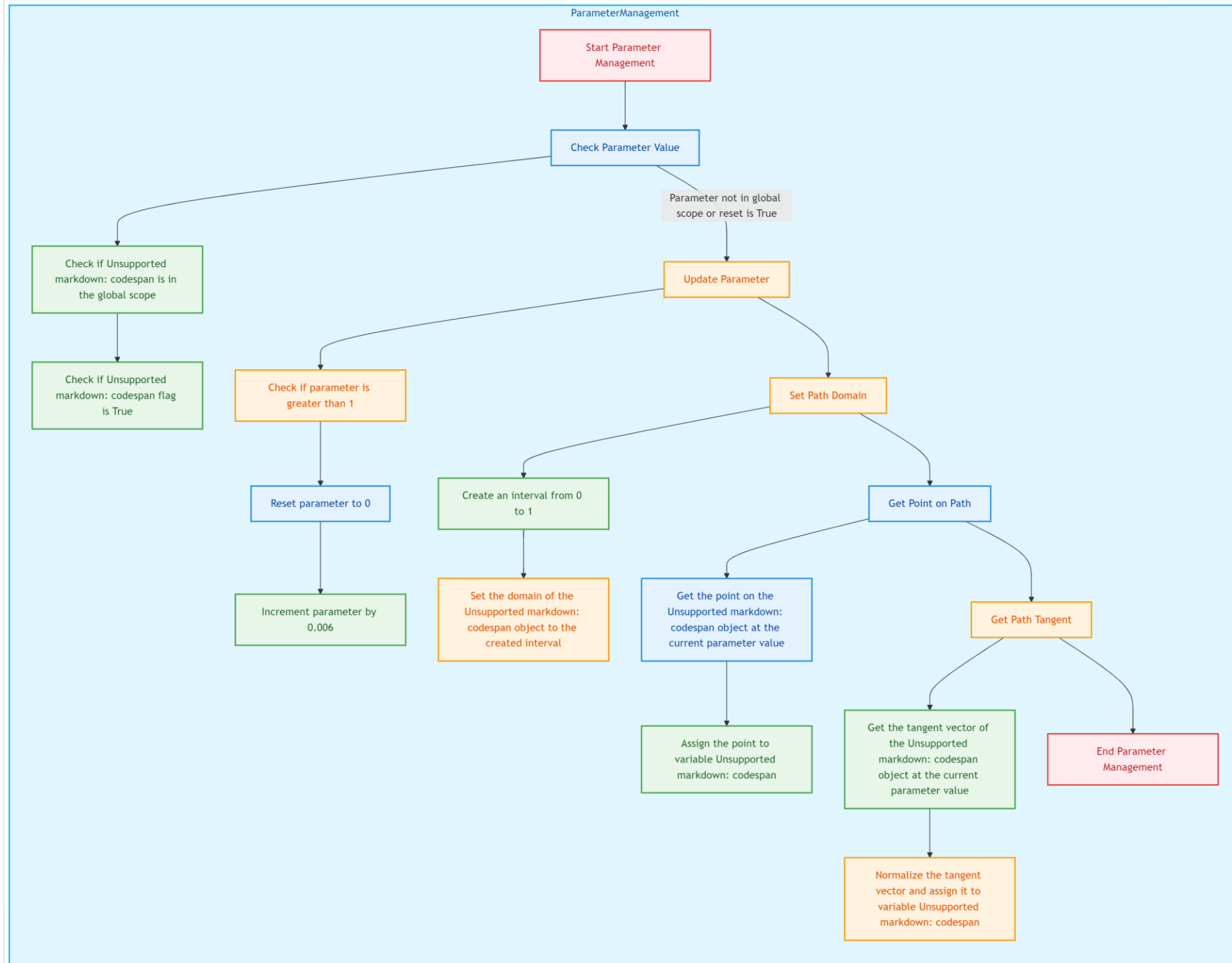
1. ParameterManagement Subgraph:

- This subgraph handles the management of the parameter variable, which is used to control the position along a path.
- **CheckParameter:** This step checks if the parameter variable is in the global scope and if the reset flag is True. If either condition is met, it proceeds to the **UpdateParameter** step.
- **UpdateParameter:** This step checks if the parameter value is greater than 1. If so, it resets the parameter to 0. Then, it increments the parameter by 0.006.
- **SetPathDomain:** This step creates an interval from 0 to 1 and sets the domain of the path object to this interval.
- **GetPointOnPath:** This step retrieves the point on the path object at the current parameter value and assigns it to the *a* variable.
- **GetPathTangent:** This step retrieves the tangent vector of the path object at the current parameter value, normalizes it, and assigns it to the *t* variable.
- **End:** This step represents the end of the ParameterManagement subgraph.

The code provided manages the parameter value, which is used to control the position along a path. It updates the parameter value, sets the domain of the path object, retrieves the point on the path at the current parameter value, and gets the tangent vector at that point. These values (*a* and *t*) can be used for further processing or visualization of the path.

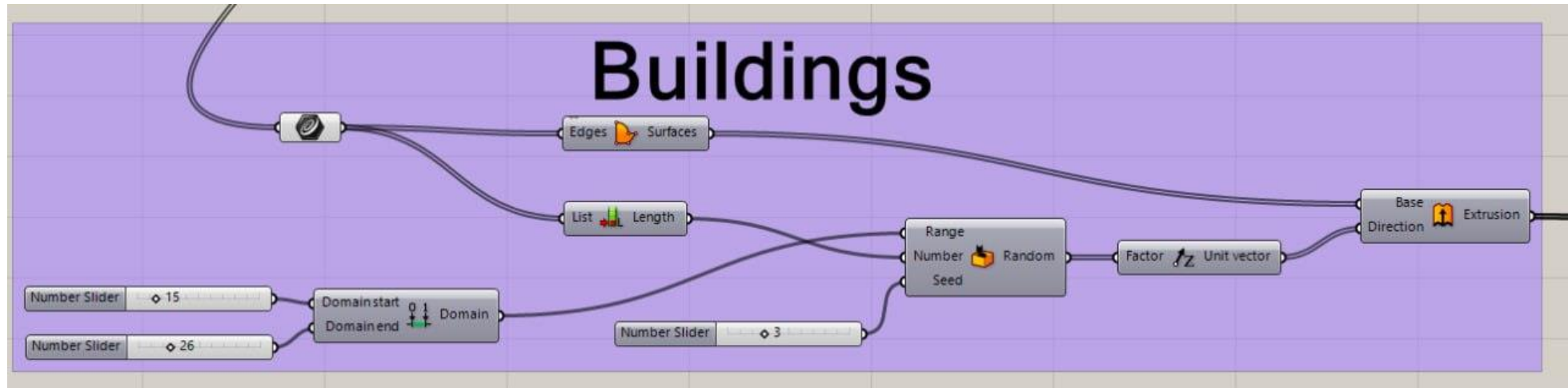
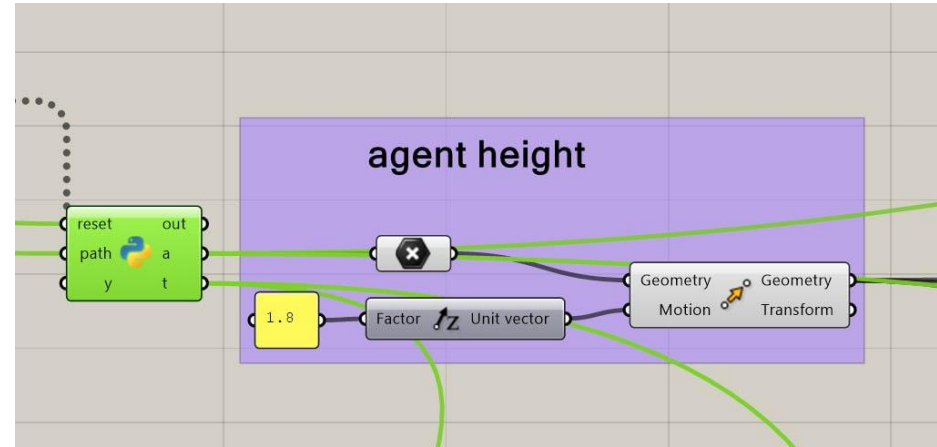


Globalization



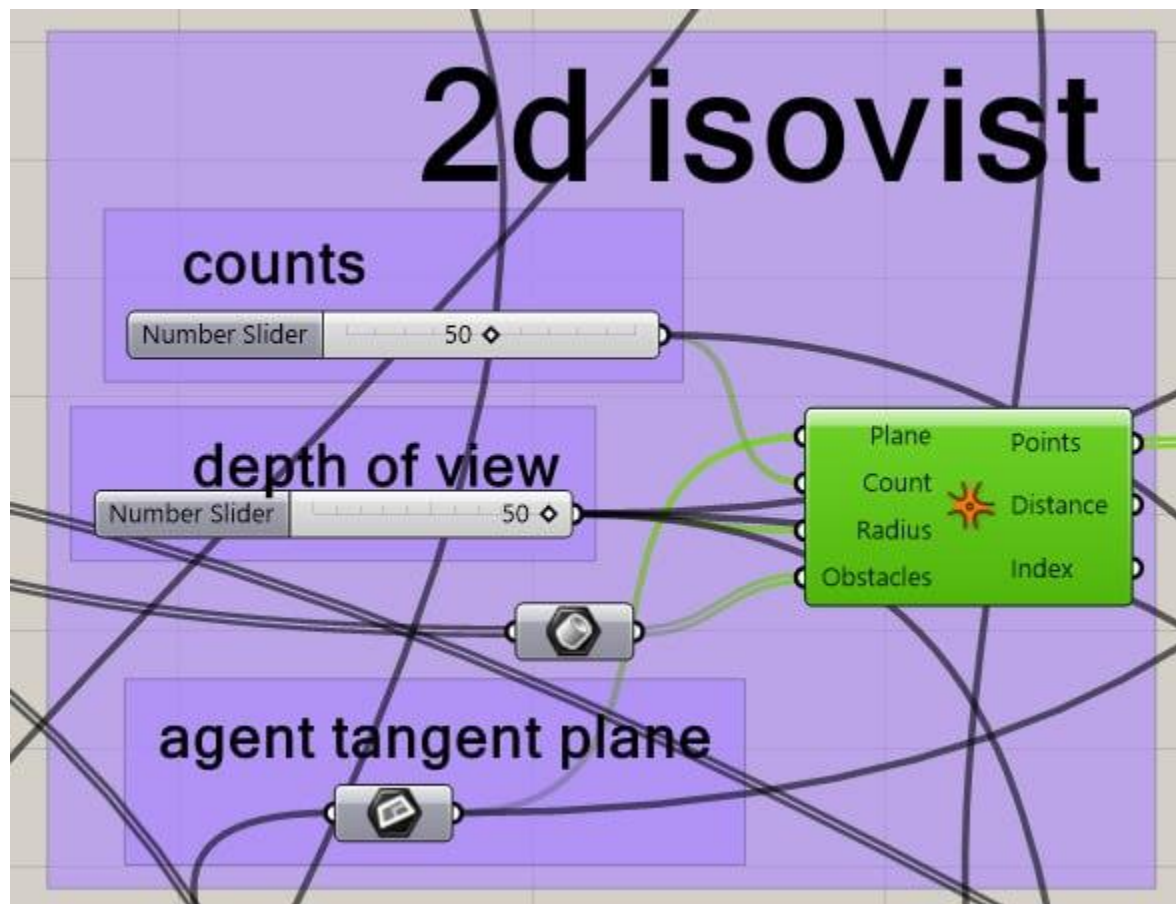


Grasshopper Codes and Algorithm





Grasshoppr Codes and Algorithm

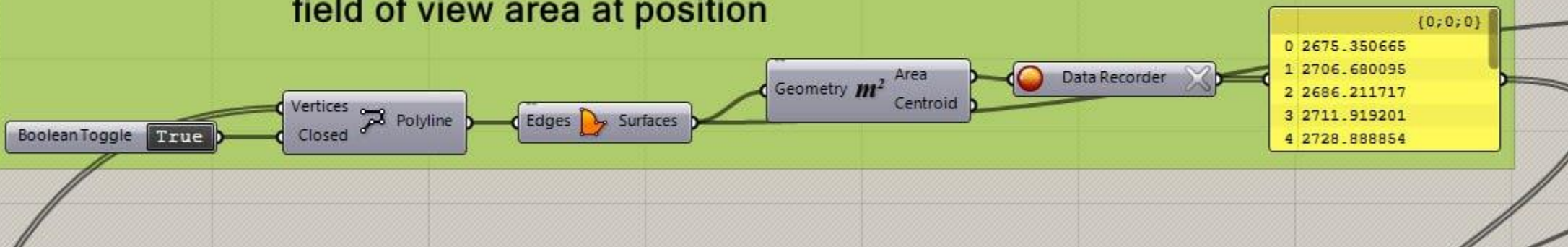




Grasshopper Codes and Algorithm

clarity of view in range

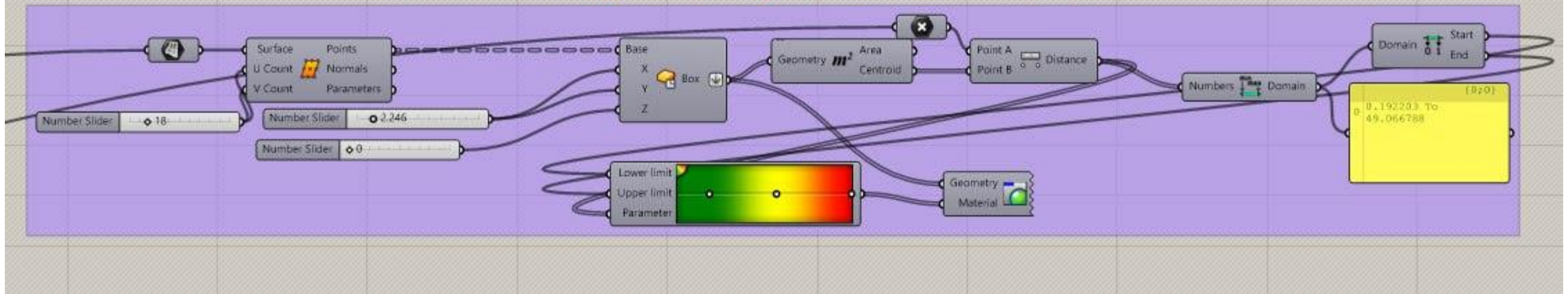
field of view area at position





Grasshopper Codes and Algorithm

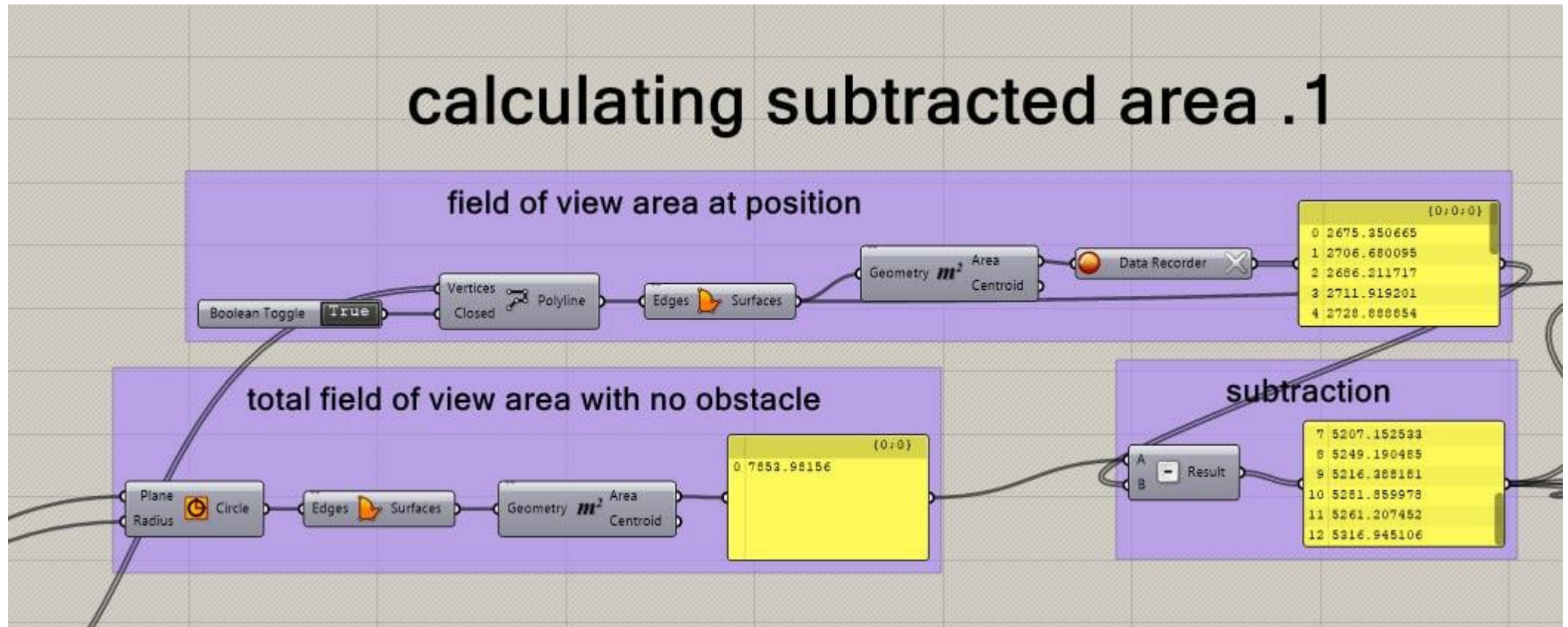
animating clarity of viewfield





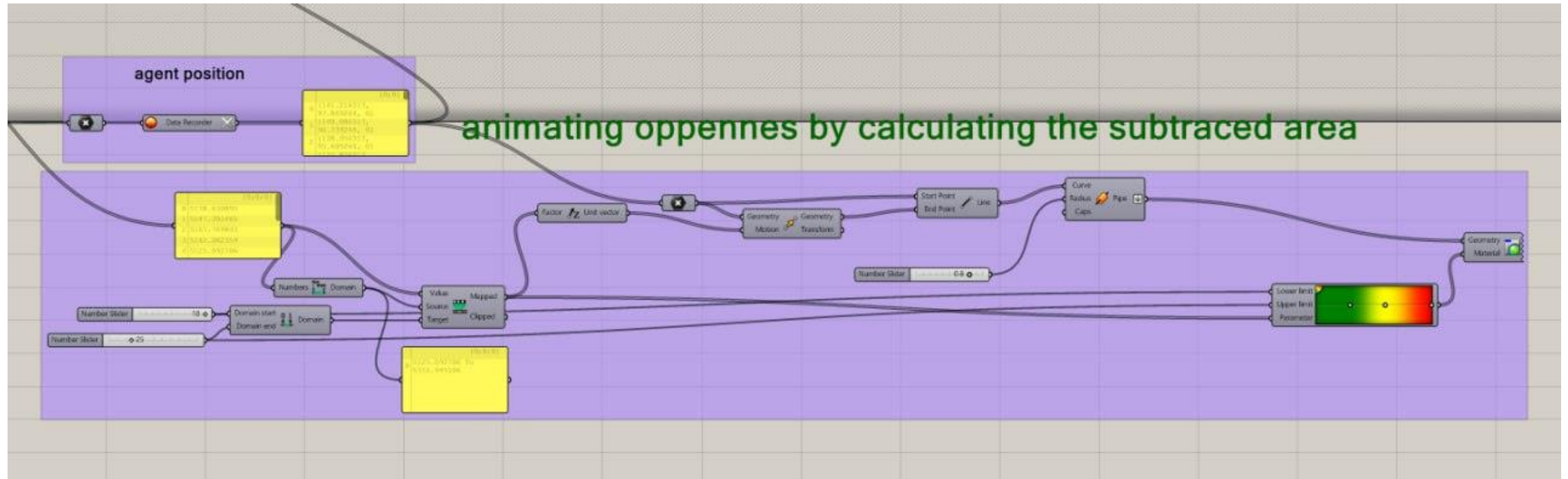
Grasshopper Codes and Algorithm

calculating subtracted area .1





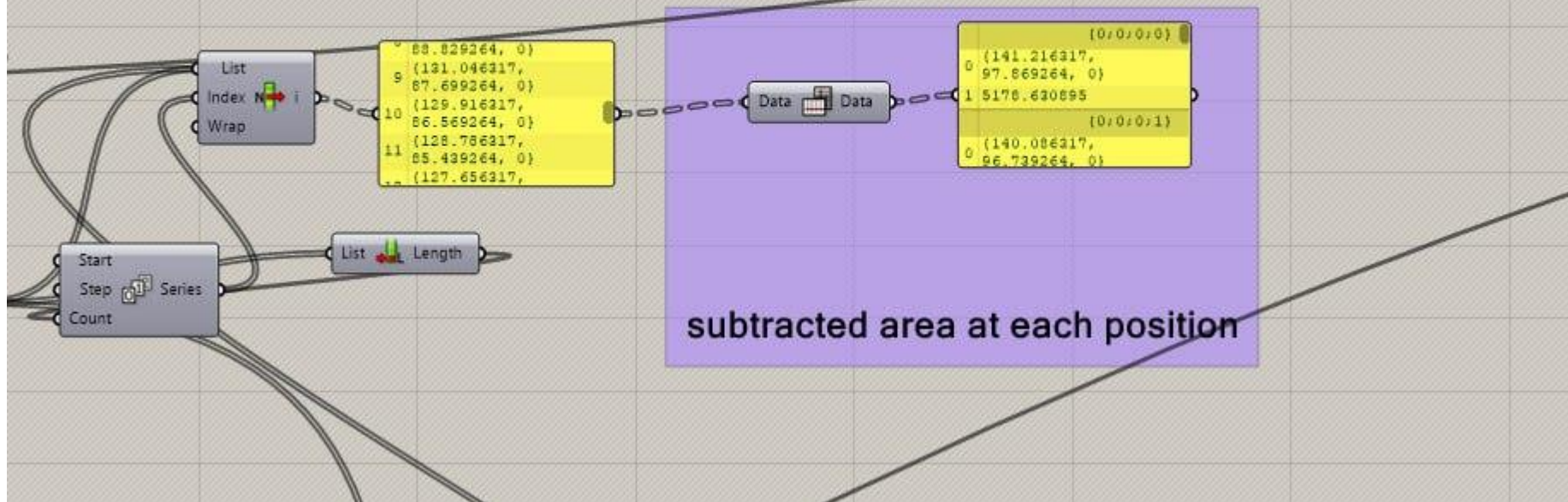
Grasshopper Codes and Algorithm





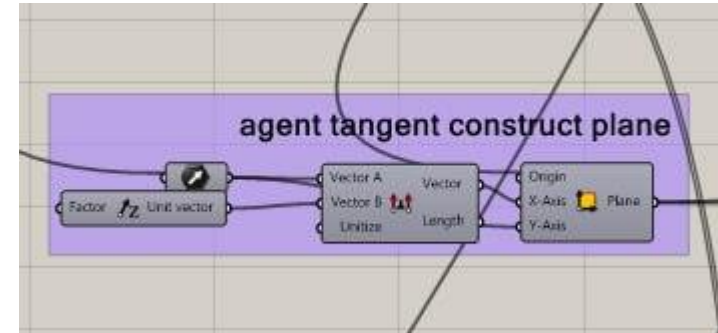
Grasshopper Codes and Algorithm

calculating subtracted area at each point

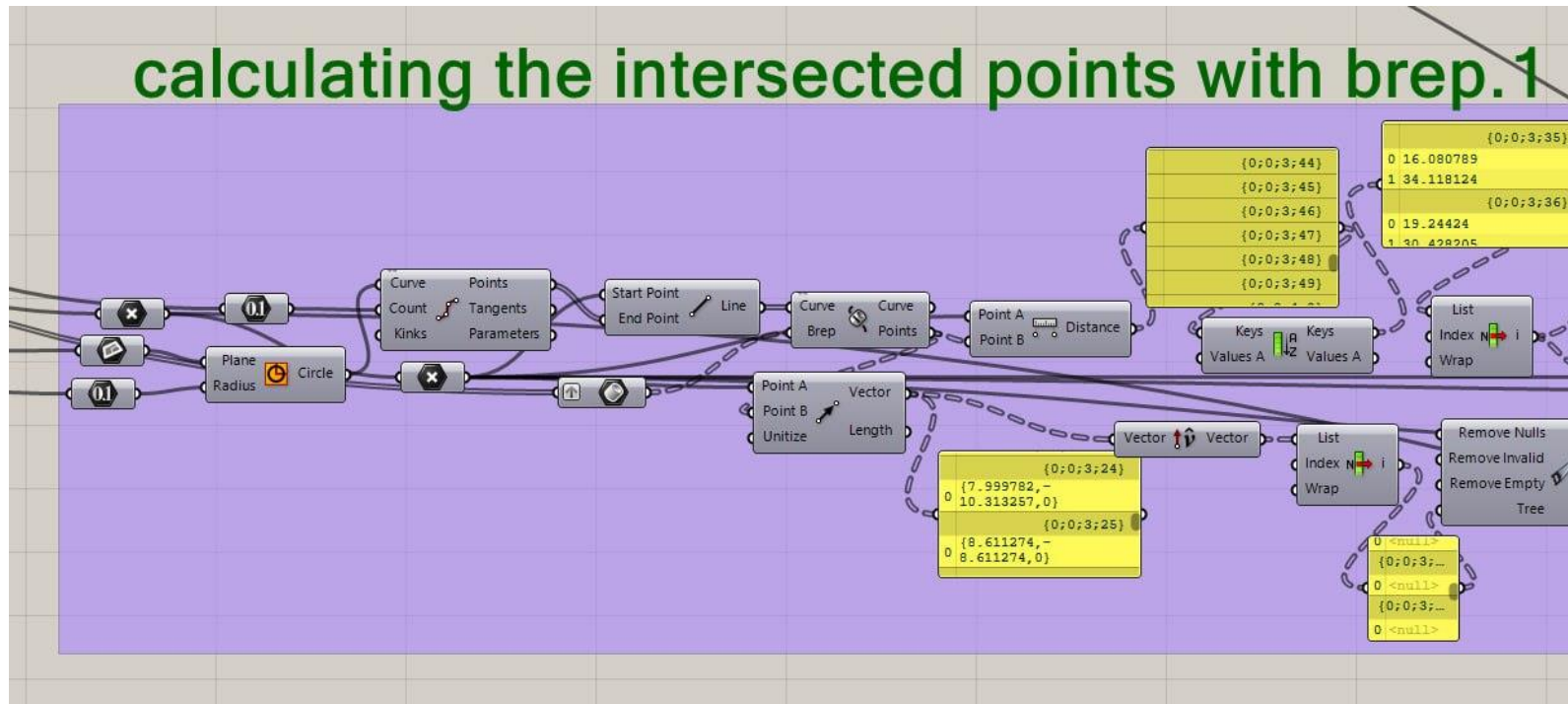




Grasshopper Codes and Algorithm



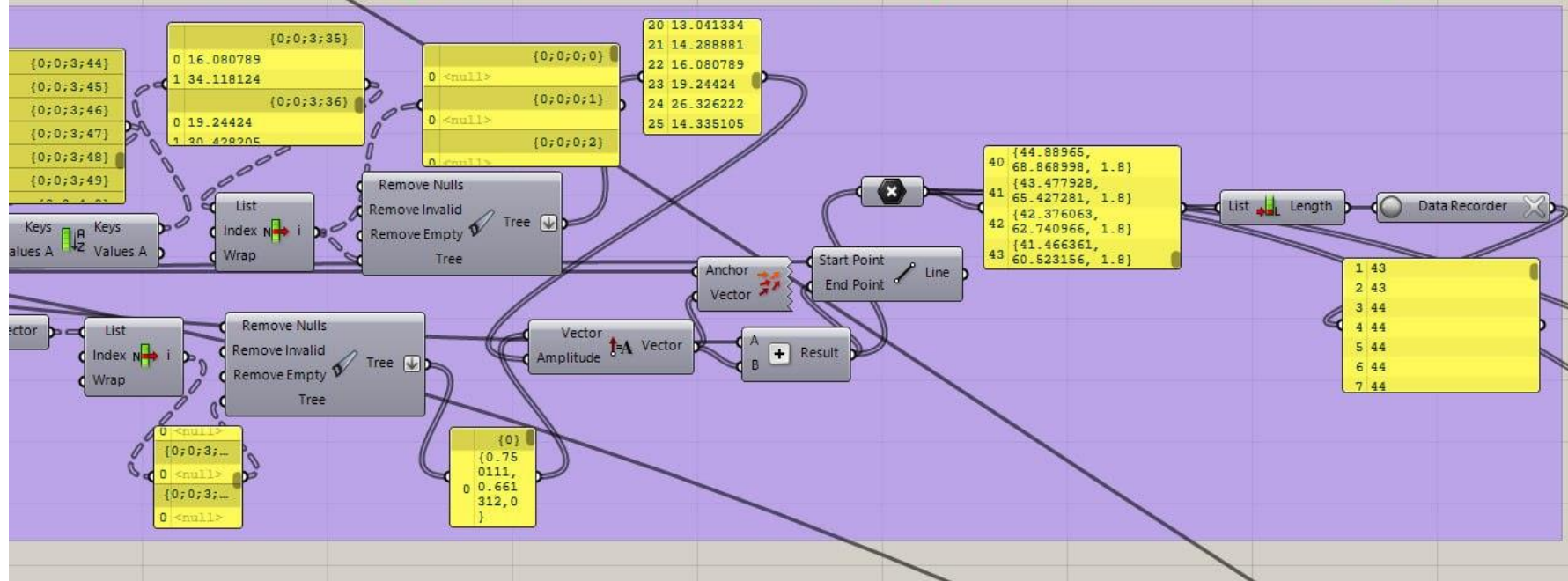
calculating the intersected points with brep.1





Grasshopper Codes and Algorithm

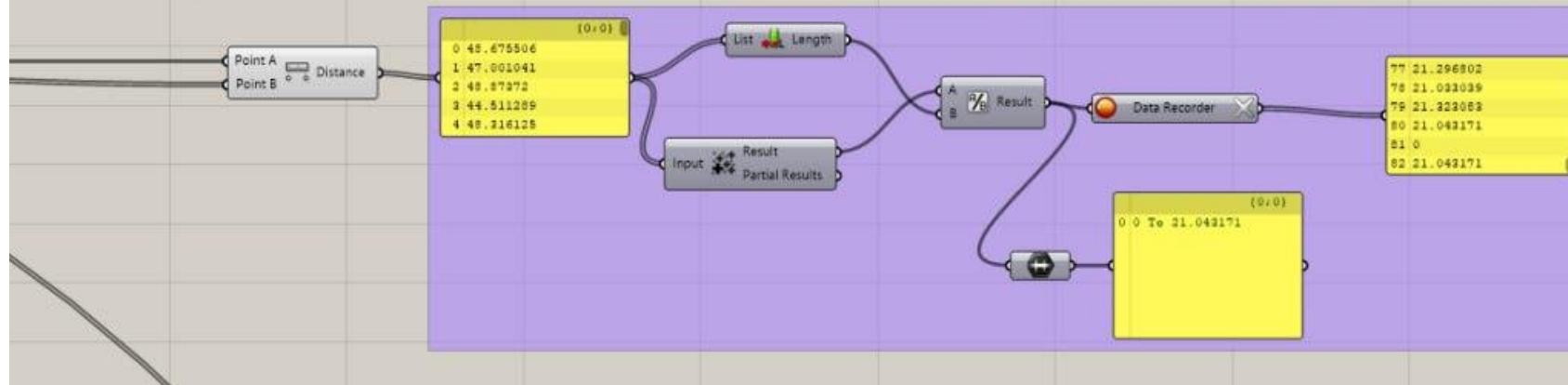
calculating the intersected points with brep.2





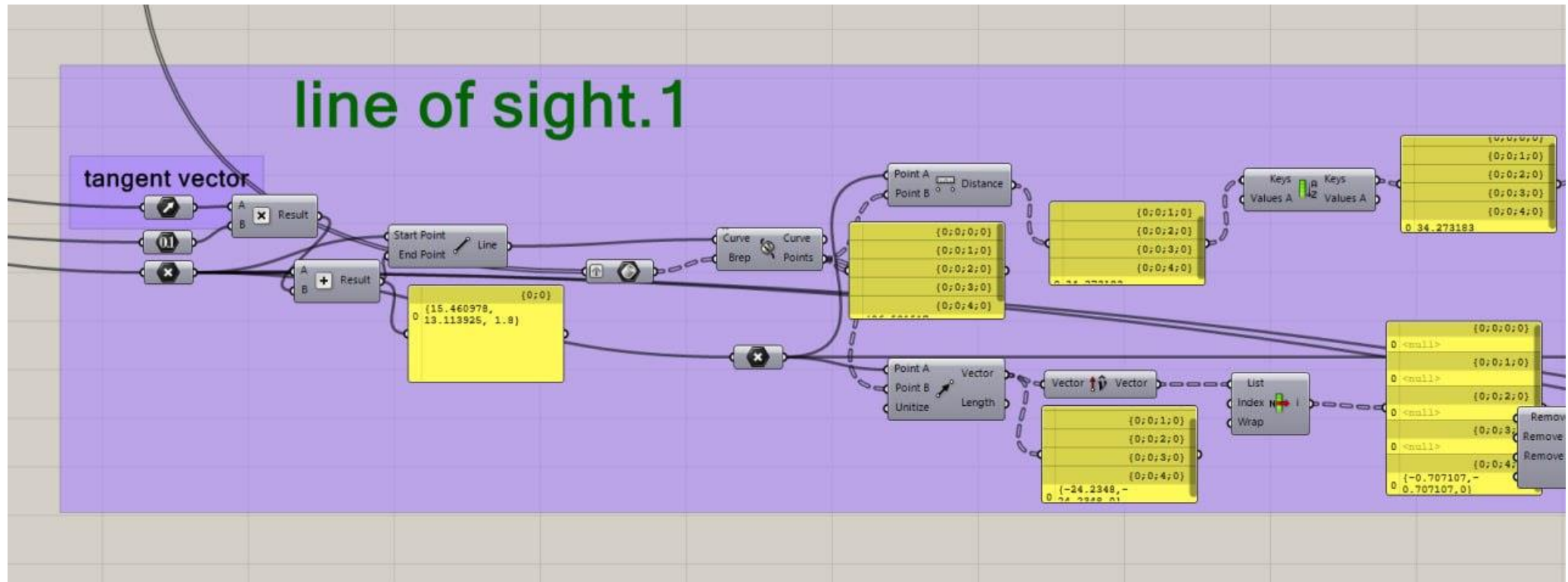
Grasshopper Codes and Algorithm

average distance from agent and intersected points(oppenness2)



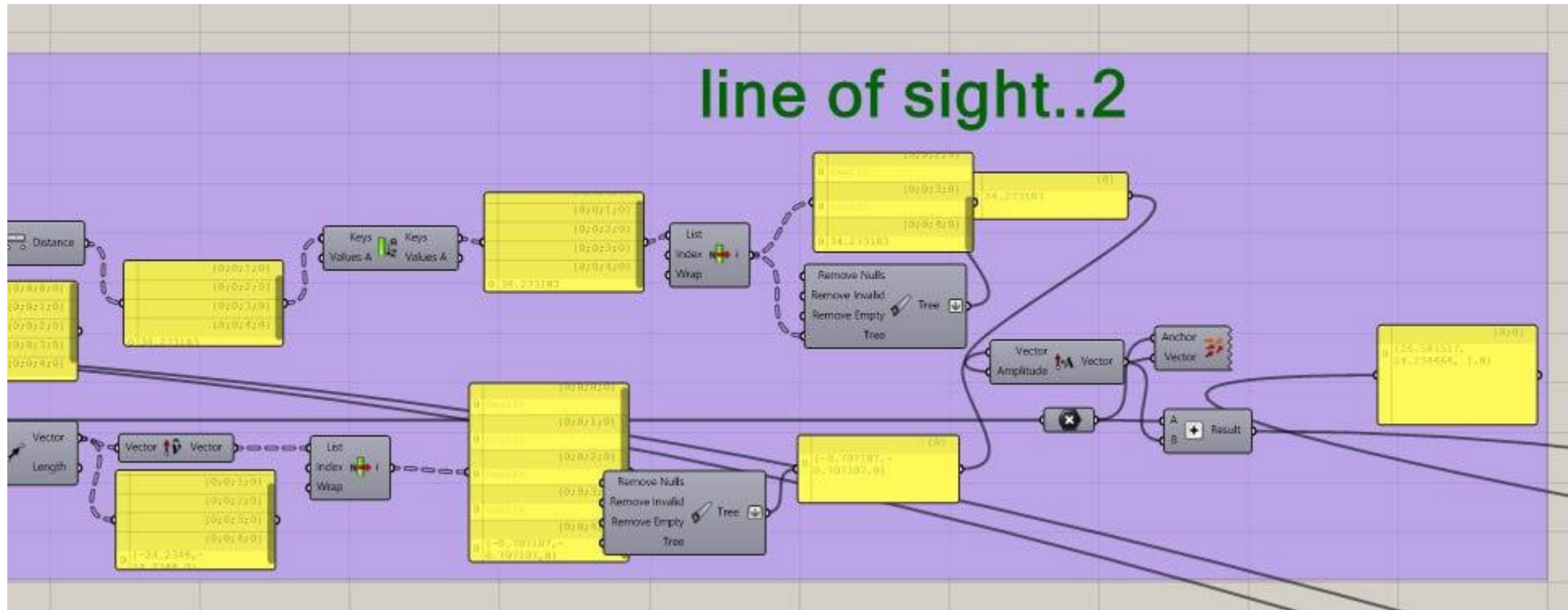


Grasshopper Codes and Algorithm



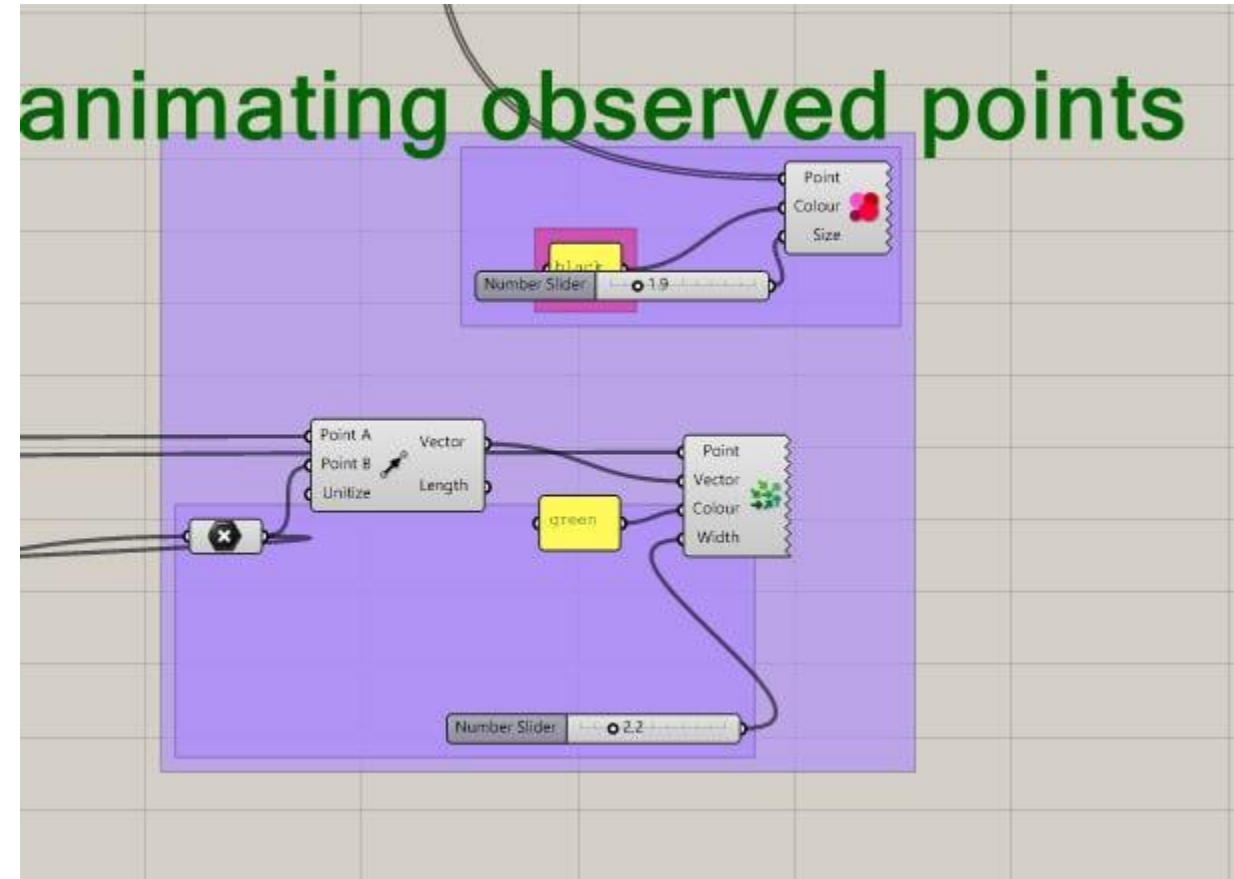


Grasshopper Codes and Algorithm





Grasshoppr Codes and Algorithm



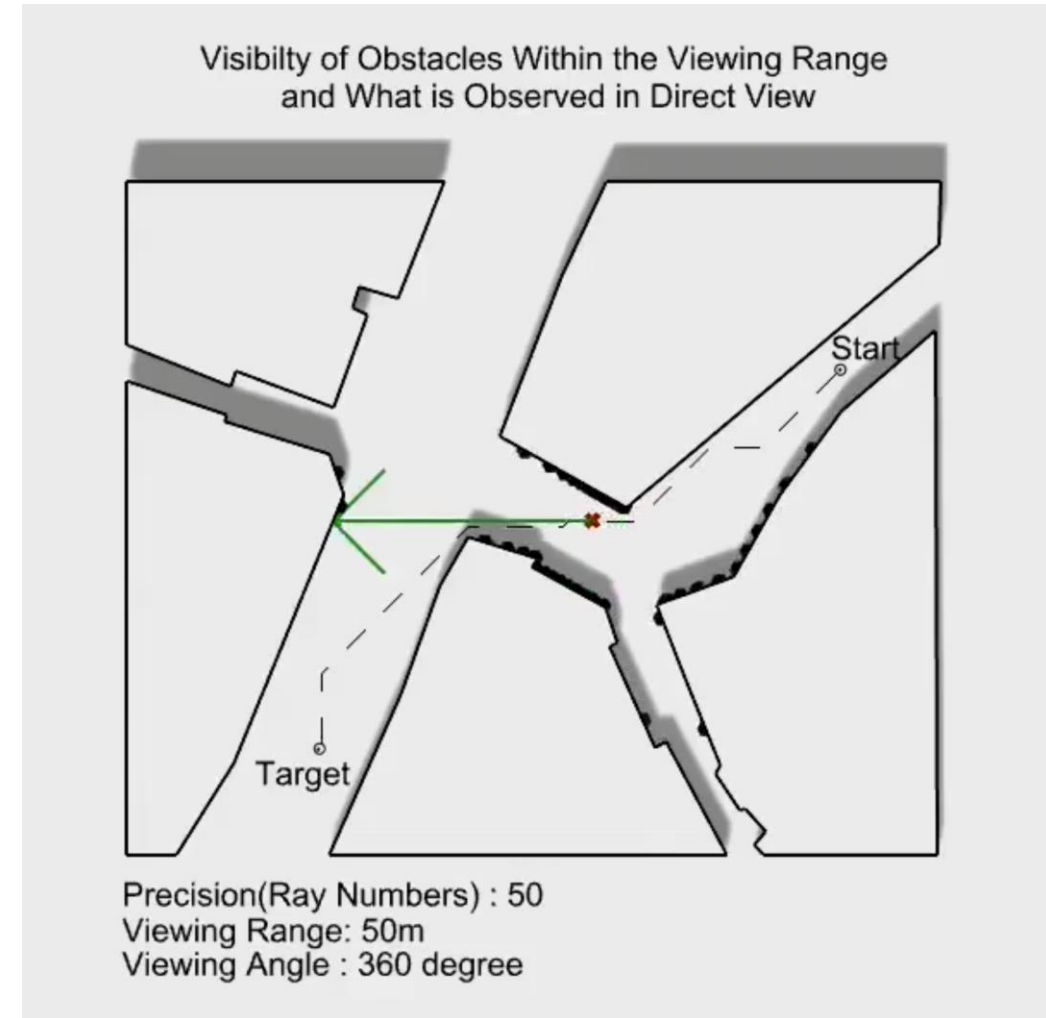
4

Output Display



Output Display

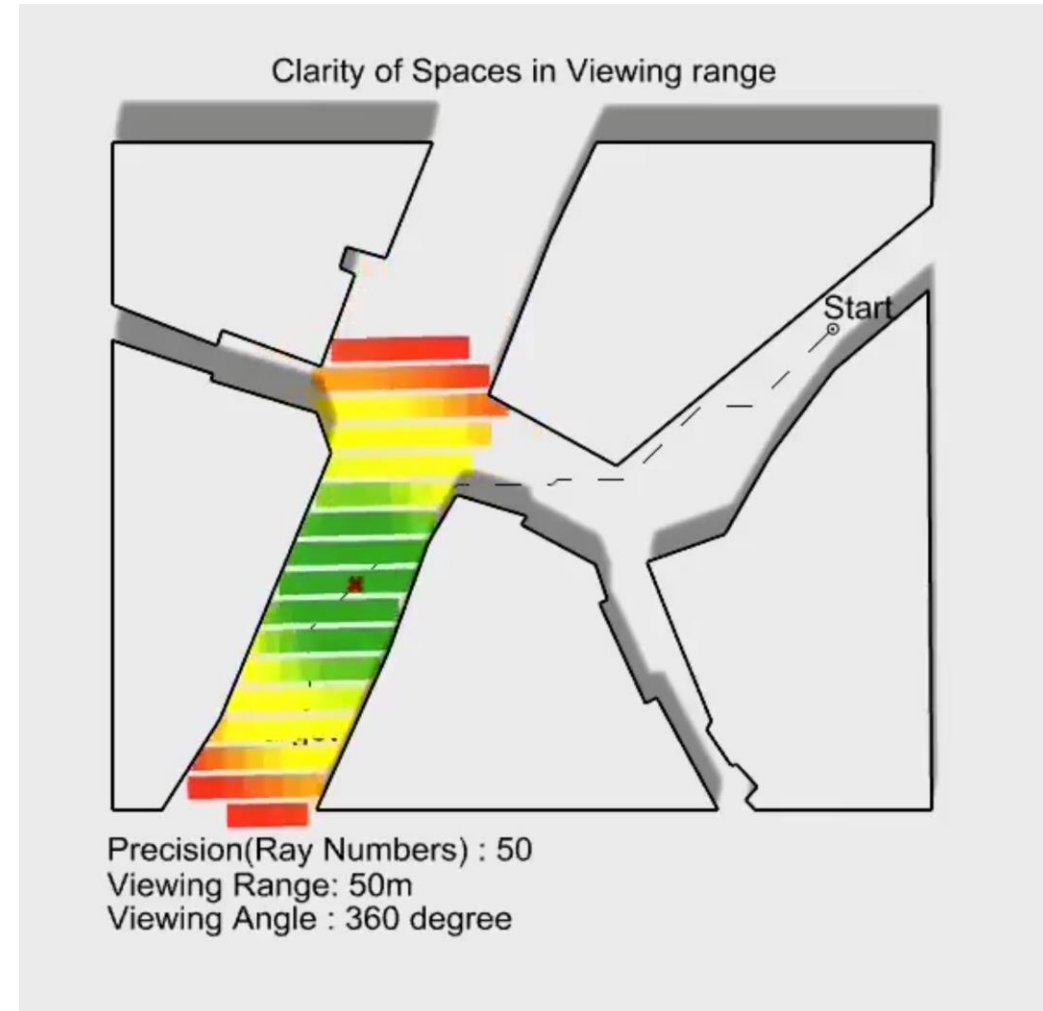
Visibility of Obstacles Within the Viewing Range and What is Observed in Direct View:





Output Display

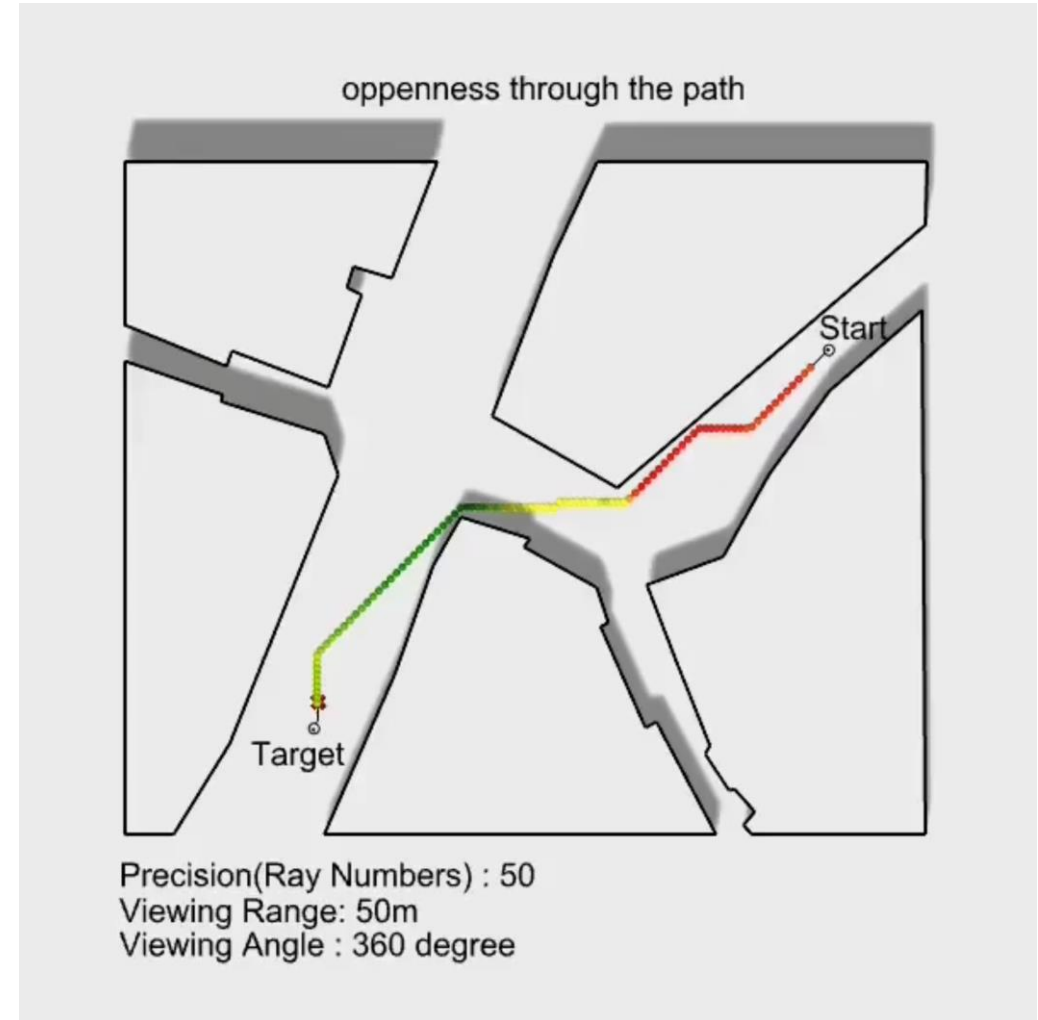
Clarity of Spaces in Viewing Range:





Output Display

Openness Through :



5

Applications in Architecture and Urban Planning



In architecture and urban planning, 2D isovist analysis is utilized to improve the design and functionality of spaces.

Spatial Layout and Design

Enhancing Visibility: Ensures key areas like entrances and waiting zones are visible and accessible, guiding the arrangement of rooms, corridors, and furniture for optimal light and sightlines.

Efficient Use of Space: Helps design layouts that avoid wasted space by pinpointing overly obstructed or overly open areas for better space utilization.

Urban Planning

Designing Streets and Public Areas: Aids in creating streets, parks, and plazas by analyzing how visible and accessible these areas are from different points, thus improving pedestrian movement and engagement.

Safety and Surveillance: Assists in placing surveillance equipment and designing sightlines to enhance public safety and reduce crime through strategic visibility planning.

Building Facade Design

Facade Design: Helps in creating facades that enhance the building's visual appeal and integrate well with the surrounding urban environment by analyzing their impact on visibility.

Accessibility and Wayfinding

Signage Location: Determines the best spots for signage and wayfinding tools by analyzing visibility from various locations within a building or urban area.

Navigation:

Improves the ease of movement through complex spaces by ensuring clear sightlines and reducing confusion.

Lighting Design

Utilizing Natural Light: Assists in designing spaces that maximize natural light by analyzing light penetration and effective usage areas.

Artificial Lighting: Guides the placement of artificial lights to ensure even illumination and avoid dark spots, particularly where visibility is important.

Human Behavior and Experience

Enhancing User Comfort: Designs spaces to be open and welcoming, ensuring users have good visual access to their surroundings.

Encouraging Interaction: Creates environments that foster social interaction by analyzing how visibility influences behavior and engagement.

6

Project Challenges

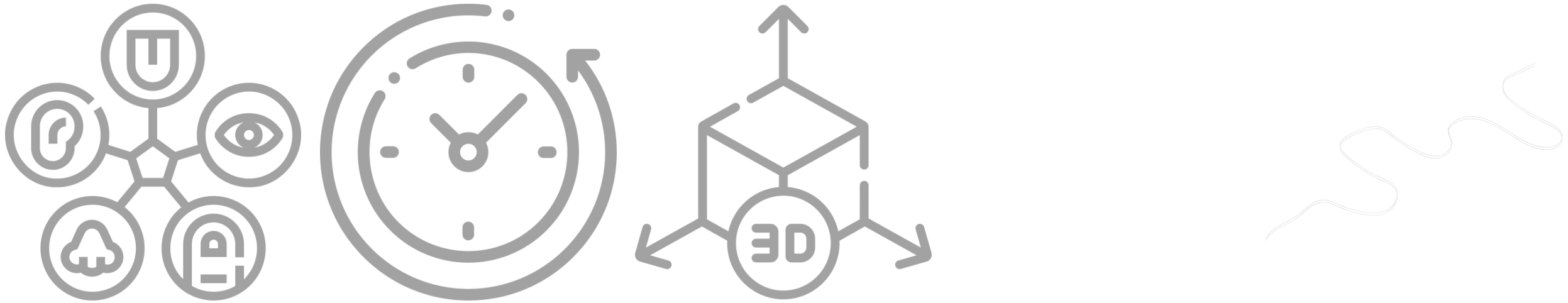


Our Challenges?

- Our main challenge in developing the algorithm was the exclusion of height as the third dimension, restricting it to a 2D framework. From the outset, we aimed to create a process that captures the user's field of view in 3D, recognizing that building heights are crucial to the visual experience. Height plays a decisive role in shaping perceptions of openness, spaciousness, and the overall sensory experience along the designed path.

7

Future Research Suggestions



This is not the end of our journey!

- One of our primary focuses is to explore the agent's sensory experiences along the path and develop new metrics accordingly. A significant challenge, as previously mentioned, is designing an algorithm that operates in three dimensions. While we've made some progress in this regard, moving forward will require collaboration with other researchers in the field. Another crucial aspect is incorporating the time dimension into the analysis, enabling us to assess how long the agent engages with each point along the path. Understanding this will greatly inform and improve our design process.